USN | | | | | | | | | | [22AIDE31]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER - 2024

FUZZY LOGIC AND APPLICATION

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			5 x2=10
1.	a.	Define uncertainty and imprecision in the context of decision-making systems.	L1	CO1	02
	b.	Name two operations that can be performed on crisp relations.	L1	CO2	02
	c.	Define defuzzification in fuzzy logic.	L1	CO3	02
	d.	Explain the concept of likelihood in fuzzy rule-based systems.	L2	CO4	02
	e.	Explain decision making under fuzzy states and fuzzy actions.	L2	CO5	02
		PART-B	6 X 5=30		
2.	a.	Explain the concept of fuzzy logic with a suitable example.	L2	CO1	05
		OR			
	b.	Describe the following terms with an example. i. Linguistic Variables ii. Uncertainty and Imprecision	L2	CO1	05
3.	a.	Demonstrate the properties of crisp relations.	L3	CO2	05
		OR			
	b.	Demonstrate the operations on fuzzy relations	L3	CO2	05
4.	a.	Employ the knowledge of fuzzy logic to arithmetic operations.	L3	CO3	05
		OR			
	b.	Illustrate defuzzification with suitable examples.	L3	CO3	05
5.	a.	Compare the different forms of fuzzy qualification.	L4	CO4	05
		OR			
	b.	Compare the performance of conjunctive and disjunctive fuzzy rules in fuzzy system.	L4	CO4	05
6.	a.	Explain the how fuzzy synthetic evaluation helps in fuzzy decision making with suitable example.	L2	CO5	05
		OR			
	b.	Describe how fuzzy ordering helps in fuzzy decision making with suitable example.	L2	CO5	05

JAIN (Deemed-to-be University) Learning Resource Centre Jain Global Campus Ramanagar-112

7.	**a.**	Illustrate the two common measures of preferences in fuzzy decision making.	L3	CO6	**05**
		OR			
	b.	Demonstrate how to Compute the Cartesian product of the two sets A ={1, 2, 3} and B = {1, 2 , 3, 4}.	L3	CO6	**05**
		PART-C		**6 X 10=60**	
8.	**a.**	Explain the architecture of fuzzy logic.	L2	CO1	**10**
		OR			
	b.	Explain the concept of random process and Describe the properties of fuzzy logic.	L2	CO1	**10**
9.	**a.**	Illustrate the terms tolerance and equivalence relations in fuzzy logic.	L3	CO2	**10**
		OR			
	b.	Interpret with suitable examples, the differences between interactive and non-interactive fuzzy sets.	L3	CO2	**10**
10.	**a.**	Consider the following fuzzy sets **A** = { (x1, 0.2), (x2, 0.5), (x3, 0.6), (x4, 0.8), (x5, 1.0) } **B** = { (x1, 0.8), (x2, 0.6), (x3, 0.4), (x4, 0.2), (x5, 0.1) }. Compute the following **i.** **C** = **A** ∪ **B** (Fuzzy Union) **ii.** **C** = **A** ∩ **B** (Fuzzy intersection) **iii.** $\mathbf{A}^c$ (Fuzzy complement)	L3	CO3	**10**
		OR			
	b.	Consider the two closed intervals I1= [-1, 3] and I2 =[1,5]. Interpret the values of the following the fuzzy arithmetic operations on closed intervals. **i.** Addition I1 + I2 **ii.** Subtraction I1 – I2 **iii.** Multiplication I1 . I2 **iv.** Division I1 / I2	L3	CO3	**10**
11.	**a.**	Explain the steps for computing the output in Mamdani fuzzy inference system.	L2	CO4	**10**
		OR			
	b.	Explain the use of graphical techniques in fuzzy inference. Discuss how graphs like help visualize fuzzy rules and the inference process, providing examples where applicable.	L2	CO4	**10**

12.	**a.**	Demonstrate the concept of multi-objective decision making in fuzzy system.	L3	CO5	**10**
		OR			
	b.	Illustrate cluster analysis and cluster validity, and compare Hard c-Means (HCM) and Fuzzy c-Means (FCM) clustering methods.	L3	CO5	**10**
13.	**a.**	Use the fuzzy similarity relation R for the fuzzy partition $\tilde{U}$ given below for two classes and three data points using clustering. Also find the number of possible ways to cluster these data points in two clusters (c=2). $$\tilde{u} = \begin{bmatrix} 0.91 & 0.58 & 0.13 \\ 0.09 & 0.42 & 0.87 \end{bmatrix}$$	L3	CO6	**10**
		OR			
	b.	Consider the following fuzzy sets P and Q.	L3	CO6	**10**

μ(x)	**x1**	**X2**	**X3**	**X4**	**X5**
P	0.1	0.2	0.7	0.5	0.4
Q	0.9	0.6	0.3	0.2	0.8

Compute the lambda-cut sets for the below.

i. $P_{0.2}$, $Q_{0.3}$

ii. $(P \cup Q)_{0.6}$

iii. $(P \cup P)_{0.8}$

iv. $(P \cap Q)_{0.4}$